

Michaela K. Shallue

Utah State University
Utah Water Research Laboratory
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Education

- PhD, Civil and Environmental Engineering**, *Utah State University* Jan 2024 – Present
River mechanics and modeling | Advisor – Colin B. Phillips
Dissertation: Disturbance driven sediment dynamics and channel response
- MSc, Geology**, *Utah State University* Aug 2021 – Dec 2023
Geoscience education | Advisor – Joel Pederson
Thesis: Place-based storm hydrology for Native Student Engagement in Geoscience
Minor: Climate Adaptation Science
- BSc, Geology**, Honors, *Northern Arizona University* Aug 2015 – May 2019
Fluvial geomorphology | Advisor – Taylor Joyal
Thesis: Hydrologic impacts of climate and land use in Canyon de Chelly, Arizona
Minors: Honors Program, Psychological Sciences

Academic Appointments

- Research and Teaching Assistant, *Utah State University* 2024 – Present
Utah Water Research Laboratory, Civil and Environmental Engineering Department
- Research and Teaching Assistant, *Utah State University* 2021 – 2023
Department of Geosciences
- Climate Adaptation Science Fellow, *Utah State University* 2021 – 2023
Ecology Center

Teaching Experience

Utah State University

Role: Graduate student | 5 courses

Environmental and Hydrologic Data Analysis and Experimentation, Spring 2026

CEE5930/6660. Teaching Assistant.

Course size: 15 graduate students, 1 undergraduate.

Fluid Mechanics, CEE3500. Teaching Assistant. Fall 2025

Course size: 69 undergraduates.

Geomorphology Lab, GEO/WATS 3600L. Instructor (2 years). Fall 2022, 2023

Course size: 24 undergraduates.

Geomorphology Lecture, GEO/WATS 3600. Teaching Assistant (2 years). Fall 2022, 2023

Course size: 24 undergraduates.

Field Camp: Silver Islands, UT, Project, GEO 5200/5210. Teaching Assistant. Summer 2023

Course size: 13 undergraduates (8 Utah State, 5 Weber State).

Guest lectures

Physical Geology (Fall 2025), GEO1110. Guest lecture on fluvial geomorphology.

Introduction to Geology (Summer 2022, 2023), GEO1010. Guest lecture on runoff processes.

Publications

Peer-Reviewed (1)

1. Masteller, C.C., Phillips, C.B., Kostynick, R.P., Castejon-Villalobos, J.F., Lopez, C.G., **Shallue, M.K.**, Bower, J., Sigman, A., (2026). Tracking the trajectory of alluvial channel adjustment reveals along-river shifts in sediment mobility. *Geophysical Research Letters*, 53, doi.org/10.1029/2026GL122483

Non-Peer-Reviewed (2)

1. Pederson, J., Tanski, N., **Shallue, M.**, 2024. Tracking the timing and dynamics of Colorado River incision through study of river terraces near Dewey, UT, *Final Report on Canyonlands Natural History Association Discovery Pool Grant*.
2. **Shallue, M.**, 2023. Building and Using a Hydrology Experiment for Place-Based Learning with Native American Students, *Utah State University MSc Thesis*.

Other (1)

1. **Shallue, M.**, 2021. Earth Notes: The Returning Rapids of Lower Cataract Canyon, *Earth Notes: KNAU News and Talk - Arizona Public Radio*.

Conference Presentations

Presentations – Total: 12* | Oral: 7 | Poster: 6

*One presentation combined both formats and is counted in both categories

- 2026** (2) **Shallue, M.**, Patterson, N., Scamardo, J., Phillips, C.B., 2026. Rethinking Flood Frequency in Dryland Streams: Hydrologic and Geomorphic Drivers of Flood Hazard. 2026 Fall Meeting, AGU, *oral presentation*. (in prep)
- Shallue, M.**, Patterson, N., Scamardo, J., Phillips, C.B., 2026. Hydrologic and Geomorphic Controls on Flooding in Dryland Riverscapes. Geological Society of America Connects 2026, *poster*. (in prep)
- 2025** (4) **Shallue, M.**, Scamardo, J., Phillips, C.B., 2025. Beyond the Annual Maximum Series: Hydrological and Geomorphic Flood Risk in Dryland Streams. 2025 Fall Meeting, AGU, *poster*.
- Shallue, M.**, Phillips, C.B., 2025. Patterns in Suspended Sediment Reveal Dynamic River Transport Response to Dam Removal. 2025 Fall Meeting, AGU, *GeoBurst (oral + poster presentations)**.
- Shallue, M.**, Pederson, J., Searle, K., 2025. Cultivating Interest in Geoscience Among Native Students with an Outdoor Hydrology Experiment. Geological Society of America 2025 Rocky Mountain Section Meeting, *oral presentation*.
- Shallue, M.**, Phillips, C.B., 2025. Dam Removal Reveals Influence of Sediment Availability on Transport Dynamics. 2025 USU Spring Runoff Conference, *poster*.
- 2024** (2) **Shallue, M.**, Phillips, C.B., 2024. Dam Removal Reveals Influence of Sediment Availability on Transport Dynamics. 2024 Fall Meeting, AGU, *poster*.
- Shallue, M.**, 2024. Dam Removal Influence on Sediment Transport. TAPDINTO-STEM Mountain Hub Convening, *oral presentation*.

- 2023** (1) **Shallue, M.**, Chavez, E., 2023. Evapotranspiration and Water Trends in the San Luis Valley. State of the Basin 2023, *oral presentation*.
- 2022** (1) **Shallue, M.**, Huntley, N., 2022. Interdisciplinary Specialization for Graduate Students Who Want to Put Science into Action: Climate Adaptation Science (CAS). Geological Society of America Connects 2022, *oral presentation*.
- 2019** (2) **Shallue, M.**, Joyal, T.J., Shaw, N., 2019. Canyon de Chelly: Watershed Modeling to Predict the Hydrological Impacts of Climate and Land Use. Student Water Symposium, Northern Arizona University, *oral presentation*.
- Shallue, M.**, Joyal, T.J., Shaw, N., 2019. Canyon de Chelly: Watershed Modeling to Predict the Hydrological Impacts of Climate and Land Use. Undergraduate Research Symposium, Northern Arizona University, *poster*.

Grants

Awarded (4) - To date: \$7,000 awarded as lead PI.

Utah State University | Role: Graduate Student | 3 grants (3 PI)

Geological Society of America, GSA, Graduate Research Student Grant 2026
[Student, \$3,000]

Geological Society of America, GSA, Graduate Research Student Grant 2022
[Student, \$2,500]

Four Corners Geological Society, Master's Degree Thesis Grant 2022
[Student, \$1,000]

Northern Arizona University | Role: Undergraduate Student | 1 grant (1 PI)

School of Earth and Sustainability, Virginia L. Korte Undergraduate Research Award 2018
[Student, \$500]

Student Travel Grants (4) - To date: \$1,250 awarded (2024 - 2026)

Awarded from: USU School of Graduate Studies, USU College of Engineering, and the Rocky Mountain Section of the Geological Society of America, to attend the AGU Fall Meeting or GSA Connects.

Honors & Awards

Farouk El-Baz Student Research Award, Geological Society of America	2026
TAPDINTO-STEM Mentorship, TAPDINTO-STEM Alliance	2023 - 2024, 2024 - 2025
Outstanding Graduate Teaching Assistant, USU Department of Geosciences	2024
J. Stewart Williams Graduate Fellowship, USU Department of Geosciences	2022, 2023
Climate Adaptation Science Fellowship, Utah State University	2021 - 2022
Outstanding Field Camp Student, NAU Geology Department	2019
1st Place Undergraduate Presentation, NAU Student Water Symposium	2019
Mary Crawley Scholarship, NAU Cline Library	2019
Lumberjack Scholars Award, Northern Arizona University	2015 - 2019

Leadership & Service

Departmental Service

Civil and Environmental Engineering Department, <i>Utah State University</i>	
Lead Student Ambassador, Utah Water Research Laboratory	Fall 2025 - Summer 2026
Student Research Symposia Judge, USU Office of Research	April 2026
Student Ambassador, Utah Water Research Laboratory	Fall 2024 - Summer 2025
Department of Geosciences, <i>Utah State University</i>	
Rock and Fossil Day Volunteer	Feb 2024, 2025
Project Co-Lead, Native American Summer Mentorship Program	Summer 2022, 2023
URGE Pod Member, USU URGE Refinement Committee	Fall 2022 - Summer 2023
Committee Member, Ecology Center Seminar Series	Spring 2022 - Spring 2023
DEI Graduate Representative, Department of Geosciences	Fall 2021 - Fall 2023
Co-Lead, Geosciences DEI Brown Bag Lunches	Spring 2023 - Fall 2023
School of Earth and Sustainability, <i>Northern Arizona University</i>	
Research Volunteer (post-bacc)	Sept 2019 - May 2020
Undergraduate Student Researcher	Aug 2018 - May 2019
Cline Library, <i>Northern Arizona University</i>	
Department Representative, Library Spaces Committee	Oct 2019 - June 2021
Student Ambassador	Aug 2018 - May 2019

Scientific Meetings

Geological Society of America	
Student Volunteer, Annual Connects Meeting	2022
Student Volunteer, Joint Section Meeting	2018, 2022

Training

CLaSH Summer Field School (Graduate Technical Short Course), <i>CLaSH</i>	July 2026
EPSP Graduate Student Mentorship Program (mentee), <i>AGU EPSP Section</i>	Fall 2025
TEACH Cultural Competence Training, <i>Utah State University</i>	Jan 2025
Empowering Teaching Excellence (ETE), <i>Utah State University</i>	
ETE Explore College Teaching Program	Jan 2024 - Present
ETE Learning circles:	
Anti-Racist Community Engagement Principles and Practices	Sept 2026
The Norton Guide to Equity-Minded Teaching	Feb 2025
"Whatever It Is, I'm Against It": Resistance to Change in Higher Education	Oct 2024
TAPDINTO-STEM Mentorship Program (mentee), <i>USU, NSF</i>	2023 - 2025
Science, Management, & Policy Exchange, <i>USU Climate Adaptation Science</i>	March 2022, 2023
Glacial Geomorphology and Volcanology Field Camp, <i>South Dakota Mines</i>	Jul/Aug 2018

Other Service and Roles

Prehistoric Museum, <i>Utah State University</i>	
Education and Paleontology Intern	July 2022 - Aug 2022
Field and Preparatory Lab Volunteer	Aug 2021 - Present
Science Moab	
Media Outreach Volunteer	Oct 2020 - Aug 2021
Sierra Club, Grand Canyon Chapter	
Student Volunteer	Oct 2020 - Sept 2021
Hopi Water Resources	
Water Data Manager	Oct 2018 - May 2019

Cline Library, Northern Arizona University
E-Resources Library Coordinator
Assembly Hall Student Assistant
Course Reserves Student Assistant
User Experience Student Assistant

Dec 2018 – June 2021
Aug 2018 – Dec 2018
May 2018 – Dec 2018
Nov 2015 – Dec 2018

Skills

Field Experience

Techniques: River surveying; modal clast counting, Wolman pebble counting; intercept vegetation surveys; differential GPS, stadia rod surveying; bedrock and surficial mapping; sampling for optically stimulated luminescence (OSL) dating; paleontological prospecting, quarrying, and field jacketing.

Locations: Canyon de Chelly, Arizona; Book Cliffs, Utah; South Fork Eel River, California; Wasatch Plateau, Utah; Big Water, Utah; Colorado River near Dewey Bridge and Glen Canyon, Utah; Bear River Mountains, Utah; Rocky Mountains near Fort Collins, Colorado.

Laboratory Experience

Techniques: OSL sample preparation; wet and dry sieving; vertebrate fossil preparation, archival vertebrate fossil jacketing.

Software Experience

Adobe Creative Suite (Illustrator, Photoshop, and InDesign); ArcGIS Pro; ArcMap 10; QGIS; Python; GRASS GIS; MATLAB.

Memberships

American Geophysical Union
Association for Women Geoscientists
Geological Society of America
National Association of Geoscience Teachers
Society for Sedimentary Geology
Utah Geological Association

- Updated September 8, 2026 -